

UNIT - III Linear Transformations

Definition:

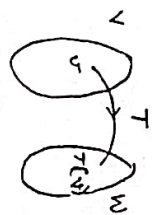
Linear Transformation: Let V & W be vector spaces over

the field F . A function $T: V \rightarrow W$ is called a Linear Transformation if

$$\forall x, y \in V \text{ & } c \in F$$

$$i) T(x+y) = T(x) + T(y)$$

$$ii) T(cx) = c T(x)$$



1) If $T: V \rightarrow W$ is a Linear Transformation, then show that $T(x-y) = T(x) - T(y)$.

Given: $T: V \rightarrow W$ is a Linear Transformation

$$i) T(x+y) = T(x) + T(y) \quad \text{ii) } T(cx) = c T(x)$$

$$\forall x, y \in V, \text{ & } c \in F$$

$$\text{L.H.S. } T(x-y) = T[x + (-y)]$$

$$\text{Since } T \text{ is a L.T.}$$

$$= T(x) + T(-y)$$

$$= T(x) + T[(-1)y] \quad [\text{By (ii)}]$$

$$= T(x) + (-1)T(y)$$

$$= T(x) - T(y) \quad \text{R.H.S.}$$

Hence proved.

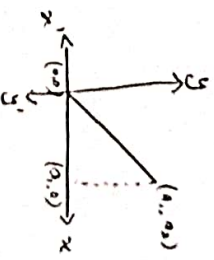
Prove that projection on x -axis is a linear transformation

Sol: Definition of projection on x -axis.

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is called the projection on x -axis.

projection is a linear:

Let $x, y \in \mathbb{R}^2, c \in \mathbb{R}$



$$\text{L.H.S. } T(cx+y) = T[c(a_1, a_2) + (b_1, b_2)]$$

$$= T[ca_1 + b_1, ca_2 + b_2]$$

$$[\text{By definition of } T]$$

$$= T(a_1, a_2) = (a_1, 0)$$

$$= (ca_1 + b_1, 0) \rightarrow \text{R.H.S.}$$

$$\text{R.H.S. } c T(x) + T(y) = c T(a_1, a_2) + T(b_1, b_2)$$

$$= c(a_1, 0) + (b_1, 0)$$

$$= (ca_1, 0) + (b_1, 0) \rightarrow \text{R.H.S.}$$

From (1) & (2), we get L.H.S. = R.H.S. a linear transformation.

Hence, projection on x -axis is a linear transformation.

3) Prove that Reflection about x -axis is a linear transformation.

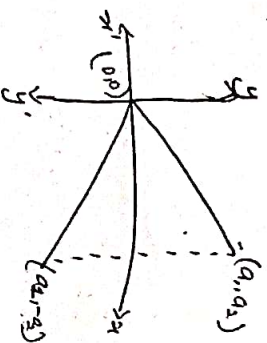
Definition of Reflection about x -axis
 Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as called the reflection about x -axis.

Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as called the reflection about x -axis.

Reflection T is linear

Let $x, y \in \mathbb{R}^2$, $c \in \mathbb{R}$

Let $x = (a_1, a_2)$, $y = (b_1, b_2)$



$$T[ax+cy] = T[c(a_1, a_2) + (b_1, b_2)]$$

$$= T(ca_1, ca_2) + (b_1, b_2)$$

$$= T(ca_1, b_1, ca_2, b_2)$$

$$= T[ca_1, b_1, ca_2, b_2] \rightarrow \text{①}$$

$$= T[ca_1, b_1, -ca_2, -b_2] \rightarrow \text{②}$$

$$L.H.S: cT(x) + T(y) = cT(a_1, a_2) + T(b_1, b_2)$$

$$= c(a_1, -a_2) + (b_1, -b_2) \quad [\text{By definition}]$$

$$= (ca_1, -ca_2) + (b_1, -b_2)$$

$$= (ca_1 + b_1, -ca_2 - b_2) \rightarrow \text{③}$$

From ① & ③, we get

$$L.H.S = R.H.S$$

Hence, Reflection about x -axis is a linear transformation.

Check if $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as given below, is a linear transformation or not.

$$a) T(a_1, a_2) = (a_1, a_2^2)$$

$$b) T(a_1, a_2) = (a_1 + 1, a_2)$$

check: T is a linear transformation or not

$$1) c) T(ax+cy) = axT(x) + cyT(y)$$

$$\forall x, y \in \mathbb{R}^2 \text{ \& } a \in \mathbb{F} = \mathbb{R}$$

$$a) \text{ Let } x = (a_1, a_2) \in \mathbb{R}^2$$

$$y = (b_1, b_2) \in \mathbb{R}^2, \alpha \in \mathbb{F} = \mathbb{R}$$

$$L.H.S: T(ax+cy) = T(\alpha(a_1, a_2) + (b_1, b_2))$$

$$= T[\alpha a_1, \alpha a_2 + (b_1, b_2)]$$

$$= T[\alpha a_1 + b_1, \alpha a_2 + b_2]$$

$$= T(a_1, a_2) = (a_1, a_2^2)$$

$$\text{R.H.S: } T(ax+cy) = T(a_1, a_2)$$

$$= (a_1, a_2^2)$$

$$= (a_1, a_2^2 + b_2^2 + 2a_2b_2) \rightarrow \text{①}$$

$$R.H.S: aT(x) + T(y) = aT(a_1, a_2) + T(b_1, b_2)$$

$$= a(a_1, a_2^2) + (b_1, b_2^2)$$

$$= (aa_1, aa_2^2) + (b_1, b_2^2)$$

$$= (aa_1, aa_2^2 + b_1, b_2^2)$$

$$= (aa_1, aa_2^2) + (b_1, b_2^2)$$

$$\bullet (\alpha a_1, b_1, \alpha a_2 + b_2) \rightarrow \textcircled{2}$$

From ① & ②, we get

L.H.S $\neq$ R.H.S

Hence, T is not a Linear Transformation.

b) Let $x = (a_1, a_2) \in \mathbb{R}^2$

$$y = (b_1, b_2) \in \mathbb{R}^2, \alpha \in \mathbb{F} = \mathbb{R}$$

L.H.S:

$$T(\alpha x + y) = T[\alpha(a_1, a_2) + (b_1, b_2)] \\ = T[\alpha a_1 + b_1, \alpha a_2 + b_2]$$

$$\text{Since } T(a_1, a_2) = (a_1 + 1, a_2) \\ = (\alpha a_1 + b_1 + 1, \alpha a_2 + b_2) \rightarrow \textcircled{3}$$

R.H.S:

$$\alpha T(x) + T(y) = \alpha T(a_1, a_2) + T(b_1, b_2)$$

$$\text{Since } T(a_1, a_2) = (a_1 + 1, a_2)$$

$$= \alpha(a_1 + 1, a_2) + (b_1 + 1, b_2)$$

$$= (\alpha a_1 + \alpha + b_1 + 1, \alpha a_2 + b_2)$$

$$= (\alpha a_1 + \alpha + b_1, \alpha a_2 + b_2)$$

From ③ & ④, we get

L.H.S $\neq$ R.H.S

Hence, T is not a Linear Transformation.

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by

$T(a_1, a_2) = (2a_1 + a_2, a_1)$, then prove that T is a Linear transformation.

To prove: $T(\alpha x + y) = \alpha T(x) + T(y)$

$\forall x, y \in \mathbb{R}^2, \alpha \in \mathbb{F} = \mathbb{R}$

Let $x = (a_1, a_2) \in \mathbb{R}^2, T$

$$y = (b_1, b_2) \in \mathbb{R}^2$$

$$T(\alpha x + y) = T[\alpha(a_1, a_2) + (b_1, b_2)]$$

$$= T[\alpha a_1 + b_1, \alpha a_2 + b_2]$$

$$= T(\alpha a_1 + b_1, \alpha a_2 + b_2)$$

$$= (2(\alpha a_1 + b_1) + (\alpha a_2 + b_2), \alpha a_1 + b_1)$$

$$= (2\alpha a_1 + 2b_1 + \alpha a_2 + b_2, \alpha a_1 + b_1) \\ \rightarrow \textcircled{3}$$

R.H.S:

$$\alpha T(x) + T(y) = \alpha(2a_1 + a_2, a_1) + (2b_1 + b_2, b_1)$$

$$= (2\alpha a_1 + \alpha a_2, \alpha a_1) + (2b_1 + b_2, b_1)$$

$$= (2\alpha a_1 + \alpha a_2 + 2b_1 + b_2, \alpha a_1 + b_1) \\ \rightarrow \textcircled{4}$$

From ③ & ④, we get

L.H.S = R.H.S.

$\therefore T$ is a Linear Transformation.

a) If $T: M_{mn}(\mathbb{R}) \rightarrow M_{n \times m}(\mathbb{R})$ is defined as $T(A) = A^T$, where A^T is the transpose of A , then prove that T is a linear transformation.

To prove: $T[\alpha A + B] = \alpha T(A) + T(B)$
 $\forall A, B \in M_{mn}(\mathbb{R}), \alpha \in \mathbb{F} = \mathbb{R}$

$$\text{Let } T(A) = A^T \xrightarrow{\text{①}} \text{②} \quad T = ()^T$$

$$T(B) = B^T$$

$$\text{L.H.S. } T(\alpha A + B) = (\alpha A + B)^T = \alpha A^T + B^T$$

$$\text{R.H.S. } [\alpha T(A) + T(B)] = \alpha T(A) + T(B) = \alpha A^T + B^T$$

Hence, $M_{mn}(\mathbb{R})$ is a linear transformation

b) If $T: P_n(\mathbb{R}) \rightarrow P_{n-1}(\mathbb{R})$ is defined as $T[f(x)] = f'(x)$, where $f'(x)$ denotes the derivative of $f(x)$, then show that T is a linear transformation.

To prove: $T[\alpha f(x) + g(x)] = \alpha T[f(x)] + T[g(x)]$

$\forall f(x), g(x) \in P_n(\mathbb{R}), \alpha \in \mathbb{F} = \mathbb{R}$

$$T[f(x)] = f'(x)$$

$$T[g(x)] = g'(x)$$

$$\begin{aligned} \text{L.H.S. } T[\alpha f(x) + g(x)] &= [\alpha f'(x) + g'(x)] \\ &= \alpha f'(x) + g'(x) \quad \text{[Distributive addition]} \\ &= \alpha T[f(x)] + T[g(x)] \quad \text{[R.H.S.]} \end{aligned}$$

Hence $P_n(\mathbb{R})$ is a linear transformation.

c) Let $V = C(\mathbb{R})$, the vector space of continuous real-valued functions on $\mathbb{R}$. Let $a, b \in \mathbb{R}, a < b$. Define: $T: V \rightarrow \mathbb{R}$ by $T(f) = \int_a^b f(t) dt$ for all $f \in V$, then prove that T is a linear transformation.

To prove: $T[\alpha f(t) + g(t)] = \alpha T[f(t)] + T[g(t)]$

$\forall f(t), g(t) \in V$
 $\alpha \in \mathbb{F} = \mathbb{R}$

$$\text{Let, } T[f(t)] = \int_a^b f(t) dt \xrightarrow{\text{①}} \text{②}$$

$$T[g(t)] = \int_a^b g(t) dt \xrightarrow{\text{③}}$$

$$\begin{aligned} \text{L.H.S. } T[\alpha f(t) + g(t)] &= \int_a^b [\alpha f(t) + g(t)] dt \\ &= \int_a^b \alpha f(t) dt + \int_a^b g(t) dt \\ &= \alpha \int_a^b f(t) dt + \int_a^b g(t) dt \\ &= \alpha T[f(t)] + T[g(t)] \quad \text{[R.H.S.]} \end{aligned}$$

Hence proved

Q/3/05
Ans: If $T: V \rightarrow W$ is a L.T. Then show that

$$T(0) = 0$$

$$\text{Let } x \in V \Rightarrow T(x) \in W$$

$$\text{consider } T(x) + 0 = T(x) \quad [\because x + 0 = x]$$

$$\Rightarrow T(x) + 0 = T(x) + T(0)$$

$$\Rightarrow T(x) + 0 = T(x) + T(0) \quad [\because T \text{ is a L.T. } T(x+y) = T(x) + T(y)]$$

$$\Rightarrow T(x) + 0 = T(x) + T(0)$$

$$\Rightarrow \boxed{T(0) = 0}$$

Hence proved.

Range & Null Space:

Let V & W be vector spaces over the field F and $T: V \rightarrow W$ be a L.T.

1) Range of T is defined as set of all images of vectors in V

$$(i.e) \boxed{R(T) = \{T(x) / x \in V\}}$$

ii) Null space of T is a set of all vectors in V whose images is

$$\text{zero} \quad \boxed{N(T) = \{x \in V / T(x) = 0\}}$$

Rank and Nullity:

Let V & W be a vector spaces over the field F and $T: V \rightarrow W$ be a L.T.

If $R(T)$ and $N(T)$ are finite dimen.

$$\begin{aligned} \text{Rank of } T & \text{ is defined as,} \\ \boxed{\text{rank}(T) = \dim[R(T)]} \\ \text{Nullity of } T & \text{ is defined as,} \\ \boxed{\text{nullity}(T) = \dim[N(T)]} \end{aligned}$$

Theorem:

Proof: Let V & W be a vector spaces over the field F and $T: V \rightarrow W$ be a L.T.

If V is finite dimensional then

$$\boxed{\text{rank}(T) + \text{nullity}(T) = \dim(V)}$$

Let is also called as Rank-Nullity Theorem [RNT].

One-one Transformation:

Let V & W be a vector spaces over the field F and $T: V \rightarrow W$ be a L.T.

If for $x, y \in V$

$$T(x) = T(y)$$

$$\Rightarrow x = y$$

$$(or) x \neq y \Rightarrow T(x) \neq T(y)$$

T is said to be one-one L.T.

Onto Transformation:

Let V & W be a vector spaces over the field F and $T: V \rightarrow W$ be a L.T.

T is said to be onto iff $\forall y \in W$

$$\exists x \in V$$

$$\exists: \boxed{T(x) = y}$$

$$\text{if } \boxed{\dim[R(T)] = \dim(W)}$$

1) Proof: Is $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1 + a_2, a_1)$ one-one?

Let $x, y \in \mathbb{R}^2$, $x = (x_1, x_2) \in \mathbb{R}^2$
 $y = (y_1, y_2) \in \mathbb{R}^2$

$$\text{Now, } T(x) = T(y)$$

$$\Rightarrow T(x_1, x_2) = T(y_1, y_2)$$

$$\Rightarrow (x_1 + x_2, x_1) = (y_1 + y_2, y_1)$$

$$\therefore T(a_1, a_2) = (a_1 + a_2, a_1)$$

Equating like terms, we get

$$x_1 + x_2 = y_1 + y_2 \rightarrow \text{---} \text{---} \text{---}$$

$$\boxed{x_1 = y_1}$$

put $x_1 = y_1$ in eqn (1),

$$y_1 + x_2 = y_1 + y_2$$

$$\Rightarrow \boxed{x_2 = y_2}$$

Hence T is one-one

$$T(x) = T(y) \Rightarrow x = y$$

Is $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$T(a_1, a_2) = (a_1 + a_2, a_1 - a_2, a_2)$$

one-one?

$$\text{Let } x, y \in \mathbb{R}^2, x = (a_1, a_2) \in \mathbb{R}^2$$

$$y = (b_1, b_2) \in \mathbb{R}^2$$

$$\text{Now, } T(x) = T(y)$$

$$\Rightarrow T(a_1, a_2) = T(b_1, b_2)$$

$$\Rightarrow (a_1 + a_2, a_1 - a_2, a_2) = (b_1 + b_2, b_1 - b_2, b_2)$$

Equating like terms, we get

$$a_1 + a_2 = b_1 + b_2 \rightarrow \text{---} \text{---} \text{---}$$

$$a_1 - a_2 = b_1 - b_2 \rightarrow \text{---} \text{---} \text{---}$$

$$a_2 = b_2 \rightarrow \text{---} \text{---} \text{---}$$

put $a_2 = b_2$ in eqn (1),

$$a_1 + b_2 = b_1 + b_2$$

$$\Rightarrow \boxed{a_1 = b_1}$$

$$\Rightarrow \boxed{x = y}$$

Hence T is one-one.

Theorem:

Let V and W be vector spaces and $T: V \rightarrow W$ be a linear transformation. Then prove that T is one-one iff and only if $N(T) = \{0\}$.

Proof: Given that

$T: V \rightarrow W$ is a linear transformation.

$$(1.e) \quad T(ax+by) = aT(x) + bT(y) \longrightarrow (3) \\ \forall x, y = v \quad \& \quad a \in \mathbb{F}$$

$$ii) \quad T \text{ is } 1-1 \longrightarrow (3)$$

$$(1.e) \quad T(x) = T(y) \Rightarrow T(x) \neq T(y)$$

$$\Rightarrow x=y \text{ (or) } x \neq y \Rightarrow T(x) \neq T(y)$$

We need to prove $N(T) = \{0\}$
Let $x \in N(T)$, then by definition of $N(T)$

$$T(x) = 0$$

$$[\because T(0) = 0]$$

$$\Rightarrow T(x) = T(0)$$

$$[\text{By eqn (3)}]$$

$$\Rightarrow x = 0$$

$$\Rightarrow 0 \in N(T)$$

$$\Rightarrow N(T) = \{0\}.$$

Part II:

Given that T is a L.T.

$$N(T) = \{0\}.$$

To prove: T is 1-1

$$\text{suppose } T(x) = T(y)$$

$$\Rightarrow T(x) - T(y) = 0$$

$$\Rightarrow T(x-y) = 0 \quad [\because T \text{ is a L.T.}]$$

$$\Rightarrow x-y \in N(T) = \{0\} \quad [\text{By definition of } N(T)]$$

$$\Rightarrow x-y = 0$$

$$\Rightarrow \boxed{x=y}$$

$$\Rightarrow T \text{ is } 1-1.$$

Hence proved //

$$(1.e) \quad T(ax+by) = aT(x) + bT(y) \longrightarrow (3) \\ \forall x, y = v \quad \& \quad a \in \mathbb{F}$$

$$ii) \quad T \text{ is } 1-1 \longrightarrow (3)$$

$$(1.e) \quad T(x) = T(y) \Rightarrow T(x) \neq T(y)$$

$$\Rightarrow x=y \text{ (or) } x \neq y \Rightarrow T(x) \neq T(y)$$

We need to prove $N(T) = \{0\}$
Let $x \in N(T)$, then by definition of $N(T)$

$$T(x) = 0$$

$$[\because T(0) = 0]$$

$$\Rightarrow T(x) = T(0)$$

$$\Rightarrow x = 0 \quad [\text{By eqn (3)}]$$

$$\Rightarrow 0 \in N(T)$$

$$\Rightarrow N(T) = \{0\}.$$

Part II:

Given that T is a L.T.

$$N(T) = \{0\}.$$

To prove: T is 1-1

$$\text{suppose } T(x) = T(y)$$

$$\Rightarrow T(x) - T(y) = 0$$

$$\Rightarrow T(x-y) = 0 \quad [\because T \text{ is a L.T.}]$$

$$\Rightarrow x-y \in N(T) = \{0\} \quad [\text{By definition of } N(T)]$$

$$\Rightarrow x-y = 0$$

$$\Rightarrow \boxed{x=y}$$

$$\Rightarrow T \text{ is } 1-1.$$

Hence proved //

theorem:
 v & w be vector spaces and

let T be linear. Then $N(T)$ & $R(T)$ are
 $T: v \rightarrow w$
 subspaces of v & w respectively.

Proof:

PART I:
 To P.T $N(T)$ is a subspace of v .

By definition
 $N(T) = \{x \in v / T(x) = 0\} \subseteq v$

We need to prove

i) $0 \in N(T)$, $c \in F$, then $cx + y \in N(T)$

ii) $x, y \in N(T)$, $c \in F$, then $cx + y \in N(T)$

By the property of linear transformation T

i) $T(0) = 0$, By definition of $N(T)$

ii) $x, y \in N(T)$, $c \in F$,
 $T(y) = 0$
 $\Rightarrow T(cx) = 0$, $T(y) = 0$

Now, $T[cx + y] = cT(x) + T(y)$

Since T is linear.
 $= 0 + 0$

By definition of $N(T)$
 $cx + y \in N(T)$

From i) & ii)
 $N(T)$ is subspace of v .

PART II:

To P.T $R(T)$ is subspace of w .

By definition

$R(T) = \{T(x) \in w / x \in v\} \subseteq w$

We need to prove

i) $0 \in R(T)$

ii) $x, y \in R(T)$, $c \in F$, then
 $cx + y \in R(T)$

Proof:

i) By property of L.T
 $T(0) = 0 \Rightarrow 0 \in R(T)$ by definition of $R(T)$

ii) $x, y \in R(T)$, $c \in F$

By definition of $R(T)$,

$x = T(u)$, for some $u \in v$
 $y = T(v)$, for some $v \in v$

We need to prove that $cx + y \in R(T)$

For that, we should find some $z \in v$
 such that $T(z) = cx + y \Rightarrow z \in v$

Proof:

$T(z) = cu + v$

$= cT(u) + T(v)$ [$\because$ By $\odot$]
 $= T(cu) + T(v)$ (since T is linear)

$T(z) = T(cu + v)$
 $\Rightarrow z = cu + v$ such that

$$T(z) = \alpha u + v \in R(T)$$

$\Rightarrow R(T)$ is subspace of V .

PROVE that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ is a linear transformation. Also find bases for $N(T)$

& $R(T)$ compute Nullity & rank of T and verify dimension theorem. Also check if T is one-one or onto.

Given: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$.
Linear transformation.

TO PROVE: T is a linear transformation.

(i.e.) $T[\alpha x + y] = \alpha T(x) + T(y)$

$\alpha, y \in \mathbb{R}^3, \alpha \in F$

Let $x = (a_1, a_2, a_3) \in \mathbb{R}^3$

$y = (b_1, b_2, b_3) \in \mathbb{R}^3$ & $\alpha \in F$

L.H.S: $T[\alpha x + y]$

$$= T[\alpha(a_1, a_2, a_3) + (b_1, b_2, b_3)]$$

$$= T(\alpha a_1 + b_1, \alpha a_2 + b_2, \alpha a_3 + b_3)$$

$$\text{since } T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$$

$$= (\alpha a_1 + b_1 - \alpha a_2 - b_2, 2\alpha a_3 + 2b_3) \rightarrow R$$

R.H.S:

$$\alpha T(x) + T(y) = \alpha T(a_1, a_2, a_3) + T(b_1, b_2, b_3)$$

$$\text{since } T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$$

$$= \alpha(a_1 - a_2, 2a_3) + (b_1 - b_2, 2b_3)$$

$$= (\alpha a_1 - \alpha a_2, 2\alpha a_3) + (b_1 - b_2, 2b_3)$$

$$= (\alpha a_1 + b_1 - \alpha a_2 - b_2, 2\alpha a_3 + 2b_3) \rightarrow R$$

From (i) we get.

$$\text{L.H.S} = \text{R.H.S}$$

$\therefore T$ is a linear transformation.

PART II

To find the basis of $N(T)$

By definition of $N(T)$

$$N(T) = \{x \in \mathbb{R}^3 / T(x) = 0\}$$

$$= \{(a_1, a_2, a_3) \in \mathbb{R}^3 / T(a_1, a_2, a_3) = (0, 0)\}$$

$$= \{(a_1, a_2, a_3) \in \mathbb{R}^3 / (a_1 - a_2, 2a_3) = (0, 0)\}$$

$$\text{but, } a_1 - a_2 = 0, 2a_3 = 0$$

$$a_1 = a_2$$

$$a_3 = 0$$

$$= \{(a_1, a_1, 0) \in \mathbb{R}^3 / a_1 \in \mathbb{R}\}$$

$$= \{a_1(1, 1, 0) / a_1 \in \mathbb{R}\}$$

$$N(T) = \text{span}\{(1, 1, 0)\}$$

$$\therefore \text{Nullity}(T) = \dim[N(T)]$$

PART III

To find the basis of $R(T)$

By definition of $R(T)$

$$R(T) = \{T(v) \in \mathbb{R}^3 / v \in \mathbb{R}^3\}$$

$$= \{T(v) / v = \alpha_1(1, 1, 0) + \alpha_2(0, 1, 0)\}$$

$$= \{T[a_1(1,0,0) + a_2(0,1,0) + a_3(0,0,1)] / a_1, a_2, a_3 \in \mathbb{R}\}$$

$$= \{a_1 T(1,0,0) + a_2 T(0,1,0) + a_3 T(0,0,1)\}$$

[Since T is L.T.]

$$\text{Since } T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$$

$$T(1,0,0) = (1,0)$$

$$T(0,1,0) = (-1,0)$$

$$T(0,0,1) = (0,2)$$

$$= a_1(1,0) + a_2(-1,0) + a_3(0,2)$$

$$= (a_1 - a_2)(1,0) + 2a_3(0,1)$$

$$\text{Span of } R(T) = \{(1,0), (0,1)\}$$

$$\boxed{\dim[R(T)] = 2} \Rightarrow \text{Rank}(T)$$

PART IV:

To verify RND Theorem [or] Dimension Theorem

$$\dim[N(T)] + \dim[R(T)] = \dim[V]$$

$$\Rightarrow 1 + 2 = 3 \quad (\because \dim(V) = \dim(\mathbb{R}^3) = 3)$$

$$\Rightarrow 3 = 3$$

Hence Dimension theorem, verified.

PART V:

To verify T is 1-1 (or) onto:

Since $N(T) \neq \{0\}$

$\therefore T$ is not one-one (By theorem)

Since $\dim[R(T)] = 2 = \dim(W) = \dim(\mathbb{R}^2)$

$\Rightarrow T$ is onto.

2) Prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)$$

is a linear transformation. Also find Basis for $N(T)$.

Compute nullity and rank of T and

verify dimension (RND) theorem. Also check

if T is one-one or onto.

Solution:

Given: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)$$

PART I:

To prove: T is a linear transformation.

$$(s, t) T[ax + ty] = sT(ax + ty) + tT(y)$$

$$\forall x, y \in \mathbb{R}^2, a \in \mathbb{F} = \mathbb{R}$$

$$\text{Let } x = (a_1, a_2)$$

$$y = (b_1, b_2)$$

$$T[ax + ty] = T[a(a_1, a_2) + t(b_1, b_2)]$$

$$= T[(aa_1, aa_2) + (tb_1, tb_2)]$$

$$= T(a_1, b_1, a_2, b_2)$$

$$\text{Since } T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)$$

$$= (a_1 + b_1 + a_2 + b_2, 0, 2(a_1 + b_1) - a_2 - b_2)$$

$$= (a_1 + a_2 + b_1 + b_2, 0, 2a_1 + 2b_1 - a_2 - b_2)$$

$$= (a_1 + a_2, 0, 2a_1 - a_2) + (b_1 + b_2, 0, 2b_1 - b_2)$$

PART II:

$$T(ax) + T(ty) = aT(a_1, a_2) + tT(b_1, b_2)$$

2) Prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)$ is a linear transformation. Also find Basis for $N(T)$ & $R(T)$. Compute nullity and rank of T and verify dimension (RND) theorem. Also check if T is one-one or onto.

Solution:

Given: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)$$

PART I:

To prove: T is a linear transformation.

$$(i.e) T[ax + y] = aT(x) + T(y)$$

$$\forall x, y \in \mathbb{R}^2, a \in \mathbb{F} = \mathbb{R}$$

$$\text{Let } x = (a_1, a_2)$$

$$y = (b_1, b_2)$$

L.H.S:

$$T[ax + y] = T[a(a_1, a_2) + (b_1, b_2)]$$

$$= T[(aa_1, aa_2) + (b_1, b_2)]$$

$$= T(aa_1 + b_1, aa_2 + b_2)$$

$$[\text{Since } T(a_1, a_2) = (a_1 + a_2, 0, 2a_1 - a_2)]$$

$$= aa_1 + b_1 + aa_2 + b_2, 0, 2(aa_1 + b_1) - aa_2 - b_2$$

$$= aa_1 + aa_2 + b_1 + b_2, 0, 2aa_1 + 2b_1 - aa_2 - b_2 \rightarrow 0$$

R.H.S:

$$aT(x) + T(y) = aT(a_1, a_2) + T(b_1, b_2)$$

From $\textcircled{1} \& \textcircled{2}$

$$\text{L.H.S} = \text{R.H.S}$$

$\therefore T$ is a linear transformation.

PART II:

To find the Basis of $N(T)$

$$N(T) = \{(a_1, a_2) \in \mathbb{R}^2 / T(a_1, a_2) = (0, 0, 0)\}$$

0 element of $\mathbb{R}^3 \Rightarrow (0, 0, 0)$

$$= \{(a_1, a_2) \in \mathbb{R}^2 / (a_1 + a_2, 0, 2a_1 - a_2) = (0, 0, 0)\}$$

$$= \{(a_1, a_2) \in \mathbb{R}^2 / a_1 = -a_2, 0, 2a_1 = a_2\} = \{(0, 0)\}$$

$$a_1 = -a_2 \rightarrow \textcircled{1}$$

$$0 = 0 \text{ (True always)} \textcircled{2}$$

$$2a_1 = a_2 \rightarrow \textcircled{3}$$

In $\textcircled{3}$, we get

$$a_1 = 0 \Rightarrow a_2 = 0$$

They are linearly independent

the Basis of $N(T)$ is $\{0\}$.

(i.e) There is no non-zero vector

$$\dim[N(T)] = \text{Nullity of } T = 0$$

$\therefore$ Basis is the empty set.

PART III

To find the Basis of $R(T)$

By definition of $R(T)$

$$R(T) = \{T(V) \mid V \in V \in \mathbb{R}^2\}$$

$$= \{T(V) \mid V = \alpha_1(1,0) + \alpha_2(1,0) + \alpha_3(1,0) + \alpha_4(1,0)\}$$

$$[\alpha_1 + \alpha_2 = 0, \alpha_3 = 0, \alpha_4 = 0]$$

$$R(T) = \text{span}\{(1,0,2), (1,0,-1)\}$$

So, they are linearly independent.

$$\dim[R(T)] = 2 = \text{rank}(T)$$

PART IV

To verify RND theorem

$$\text{Rank}(T) + \text{Nullity of } T = \dim(\mathbb{R}^2)$$

$$2 + 0 = 2$$

$$2 = 2$$

Hence, dimension theorem is verified.

PART V

To verify T is one-one (or) onto

$$\text{Since } \dim[N(T)] = 0 \Leftrightarrow N(T) = \{0, 0\}$$

$\therefore T$ is one-one

$$\text{But } R(T) \neq \dim(\mathbb{R}^3) = \dim(\mathbb{R}^3)$$

$$2 \neq 3$$

Hence T is not onto.

Homework

- Prove that $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a, b) = (a+b, a-b, b)$ is a linear transformation. Also find basis for $N(T)$ & $R(T)$. Compute Nullity & rank of T. and verify dimension (RND) theorem. Also check if T is one-one or onto.

Solution: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$T(a, b) = (a+b, a-b, b)$$

PROVE: T is a linear transformation.

To prove: T is a linear transformation.

$$T(ax+by) = aT(x) + bT(y)$$

$$\forall x, y \in \mathbb{R}^2, a, b \in \mathbb{R}$$

$$\text{Let } x = (a_1, b_1), y = (a_2, b_2)$$

$$T(ax+by) = T[a(a_1, b_1) + b(a_2, b_2)]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

$$= T[aa_1 + ba_2, ab_1 + bb_2] = T[aa_1 + ba_2, ab_1 + bb_2]$$

PART II:

To find Basis of $N(T)$

$$N(T) = \{ (a, b) \in \mathbb{R}^2 \mid T(a, b) = (0, 0) \}$$

0 element of $\mathbb{R}^3(0, 0, 0)$

$$= \{ (a, b) \in \mathbb{R}^2 \mid (a+b, a-b, b) = (0, 0, 0) \}$$

$$= \{ (a, b) \in \mathbb{R}^2 \mid \begin{cases} a+b=0 \\ a-b=0 \\ b=0 \end{cases} \}$$

$$a=b \rightarrow \textcircled{2}$$

$$b=0 \rightarrow \textcircled{3}$$

$$a=0$$

$$b=0$$

$$N(T) = \{0\}$$

They are linearly Independent

The Basis of $N(T)$ is $\{0\}$

Nullity of $N(T) = \dim \{0\}$

$$\dim[N(T)] = 0$$

PART III:

To find Basis of $R(T)$

definition of $R(T)$ is:

$$N(T) = \{ T(v) \in \mathbb{R}^3 \mid v \in \mathbb{R}^2 \}$$

$$R(T) = \{ T(v) \in \mathbb{R}^3 \mid v \in \mathbb{R}^2 \}$$

$$R(T) = \{ T(v) \in \mathbb{R}^3 \mid v = \alpha_1(0, 1) + \alpha_2(1, 0) \}$$

$$R(T) = \{ T[\alpha_1(0, 1) + \alpha_2(1, 0)] \}$$

$$= \{ \alpha_1(1, -1, 1) + \alpha_2(1, 1, 0) \}$$

$$\alpha_1 + \alpha_2$$

$$\therefore \alpha_1 + \alpha_2 = 0$$

$$\alpha_1 - \alpha_2 = 0$$

$$\text{rank of } R(T) = \dim \{ (1, -1, 1), (1, 1, 0) \}$$

$$\text{rank of } R(T) = \dim[R(T)]$$

$$= 2$$

So they are linearly Independent.

PART IV:

To prove Rank Nullity dimension Theorem.

$$\text{Nullity of } N(T) + \text{Rank of } R(T) = \dim(V)$$

$$N(T)$$

$$0 + 2 = \mathbb{R}^2$$

$$\alpha = 2$$

Hence proved. Rank Nullity dimension Theorem.

PART V:

T is one-one,

$$N(T) = \{0\}$$

T is onto,

$$\text{Rank of } R(T) = \dim(W)$$

$$= \dim(\mathbb{R}^3)$$

$$2 \neq 3$$

Hence T is not onto.

4) Prove that T defined as below is a linear transformation. also find basis for $N(T)$ and $R(T)$. Compute Nullity and Rank of T and verify RND theorem. Also check if T is one-one and onto.

Define: $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ by
 $T[f(x)] = x f(x) + f'(x)$

Standard basis for $P_2 = \{1, x, x^2\}$
 $P_3 = \{1, x, x^2, x^3\}$

Solution:

Given:

Define: $T: P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$

by $T[f(x)] = x f(x) + f'(x)$

PART I:

To Prove: T is a l.t.
 i.e. $T(\alpha f(x) + \beta g(x)) = \alpha T[f(x)] + \beta T[g(x)]$
 $\forall f(x), g(x) \in P_2(\mathbb{R})$
 $\alpha \in \mathbb{F}$

Let $T[g(x)] = x g(x) + g'(x)$

$T[g(x)] = x g(x) + g'(x)$

L.H.S:

$T[\alpha f(x) + \beta g(x)] = \alpha [x f(x) + g'(x)] + [\alpha f(x) + \beta g(x)]'$

$= \alpha [x f(x) + g(x)] + \alpha f'(x) + g'(x)$

$= \alpha [x f(x) + f'(x)] + [x g(x) + g'(x)]$

$= \alpha T[f(x)] + T[g(x)] \text{ R.H.S}$

$\therefore \text{L.H.S} = \text{R.H.S}$

$\therefore T$ is a l.t

PART II.
 To find basis for $N(T)$
 By definition of $N(T)$

$N(T) = \{x \in V / T(x) = 0\}$

$= \{g \in P_2(\mathbb{R}) / T(g) = 0\}$

$= \{a + bx + cx^2 / T(a + bx + cx^2) = 0\}$

Now, $T(a + bx + cx^2) = 0$
 $\Rightarrow x(a + bx + cx^2) + (0 + b + 2cx) = 0$

$\Rightarrow ax + bx^2 + cx^3 + b + 2cx = 0$

$\Rightarrow cx^3 + bx^2 + (a + 2c)x + b = 0x^3 + 0x^2 + 0x + 0$

Equating like terms, we get

$b = 0, \quad c = 0 \quad a + 2c = 0$

$a = 0 \quad [\because c = 0]$

$\therefore N(T) = \{0\}$

$\therefore \text{span}\{N(T)\} = \{0\}$

$\Rightarrow \text{Nullity}(T) = \dim[N(T)] = 0$

PART III

To find basis for $R(T)$

By definition of $R(T)$

$R(T) = \{T(x) \in W / x \in V\}$

$= \{T[h(x)] \in P_3(\mathbb{R}) / h(x) \in P_2(\mathbb{R})\}$

We may write it as

$T[h(x)] = T(a + bx + cx^2)$

$$= x(ax + bx + cx^2) + (0 + b + 2cx) \\ = cx^3 + bx^2 + (a + 2c)x + b \\ = cx^3 + b(x^2 + 1) + (a + 2c)x$$

$$R(T) = \{cx^3 + b(1+x^2) + (a+2c)x \mid a, b, c \in \mathbb{R}\}$$

$$\text{span}[R(T)] = \{x^3, (1+x^2), x\}$$

$$\Rightarrow \boxed{\text{Rank}(T) = \dim[R(T)] = 3}$$

$$\dim(V) = \dim[P_3(\mathbb{R})] = 3$$

PART IV:

To verify RND Theorem

$$\dim[R(T)] + \dim[N(T)] = \dim(V)$$

$$\boxed{3 + 0 = 3}$$

Hence verified RND.

PART V:

To check 1-1 (or) onto.

$$\text{Since } N(T) = \{0\}$$

$$[\because N(T) = \{0\} \text{ by Th 1-1}]$$

By theorem

$$T \text{ is 1-1}$$

Here

$$\dim[R(T)] = 3 \quad \therefore \dim[P_3(\mathbb{R})] = 4$$

$$\therefore \dim[R(T)] \neq \dim[P_3(\mathbb{R})]$$

Hence T is not onto.

Prove that $T: M_{3 \times 3}(\mathbb{R}) \rightarrow M_{3 \times 3}(\mathbb{R})$ defined by $T \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix}$ is a linear transformation. Also find basis for $N(T)$.

Linear Transformation. Also find basis for $N(T)$ & $\text{Null}(T)$ and verify RND theorem. Also check if T is one-one or onto.

Given: $T: M_{3 \times 3}(\mathbb{R}) \rightarrow M_{3 \times 3}(\mathbb{R})$

$$\text{defined by } T \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix}$$

To prove: T is a linear transformation.

$$T[A+B] = T(A) + T(B)$$

$$\forall A, B \in M_{3 \times 3}(\mathbb{R})$$

$$\text{Let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}; B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

R.H.S:

$$T(A) + T(B) = T \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} + T \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

$$= \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2b_{11} - b_{12} & b_{13} + 2b_{12} \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2a_{11} - a_{12} + 2b_{11} - b_{12} & a_{13} + 2a_{12} + b_{13} + 2b_{12} \\ 0 & 0 \end{bmatrix}$$

L.H.S:

$$T[A+B] = T \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

$$= \begin{bmatrix} 2a_{11} + 2b_{11} - a_{12} - b_{12} \\ 0 \end{bmatrix}$$

$$a_{13} + b_{13} + 2a_{12} + b_{12}$$

From (1) & (2) we get

$$L.H.S = R.H.S$$

Hence, T is a Linear Transformation.

To find Basis of $N(T)$:

$$N(T) = \{A \in M_{2 \times 3}(\mathbb{R}) / T(A) = 0\}$$

$$= \left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \in M_{2 \times 3}(\mathbb{R}) / \begin{bmatrix} 2a_{11} - a_{12} - a_{13} \\ a_{13} + 2a_{12} + a_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

We may write it as,

$$2a_{11} - a_{12} = 0 \Rightarrow 2a_{11} = a_{12}$$

$$a_{13} + 2a_{12} = 0 \Rightarrow a_{13} = -2a_{12}$$

$$\text{Since } a_{12} = 2a_{11} \Rightarrow a_{13} = -2(2a_{11})$$

$$a_{13} = -4a_{11}$$

$$a_{13} = -4a_{11}$$

$$= \left\{ \begin{bmatrix} a_{11} & 2a_{11} & -4a_{11} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \in M_{2 \times 3}(\mathbb{R}) \right\}$$

Now,

$$\begin{bmatrix} a_{11} & 2a_{11} & -4a_{11} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = a_{11} \begin{bmatrix} 1 & 2 & -4 \\ 0 & 0 & 0 \end{bmatrix} + a_{21} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} + a_{22} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} + a_{23} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Span}[N(T)] = \left\{ \begin{bmatrix} 1 & 2 & -4 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

$$\dim[N(T)] = 4$$

$$\text{Nullity}(T) = 4$$

To find Basis for $R(T)$

$$R(T) = \left\{ T(A) \in M_{2 \times 3}(\mathbb{R}) / A \in M_{2 \times 3}(\mathbb{R}) \right\}$$

$$= \left\{ \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix} \in M_{2 \times 3}(\mathbb{R}) \right\}$$

We may write it as,

$$\begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix} = (2a_{11} - a_{12}) \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + (a_{13} + 2a_{12}) \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \text{Span}[R(T)] = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\}$$

$$\dim[R(T)] = 2$$

$$\text{Rank}(T) = 2$$

To verify RND

$$\text{Rank}(T) + \text{Nullity}(T) = \dim[M_{2 \times 3}(\mathbb{R})]$$

$$2 + 4 = 6$$

$$b = 6$$

Hence verified RND theorem.

To Verify $\dim$ or rank

$$\text{But } N(T) \neq \{0\}$$

T is not one-one

$$\text{Since } \dim[R(T)] = 2$$

$$\dim[M_{2 \times 3}(\mathbb{R})] = 6$$

$$\dim[R(T)] \neq \dim[M_{2 \times 3}(\mathbb{R})]$$

$\therefore T$ is not onto

Summary:

* Introduction to Linear Transformation (1y)

8m * Definition of LT

8m * Definition: Projection on x-axis
85m * Reflection about x-axis.

8m * Problems on LT (5)

2m * Definition: Basis, $N(T)$, $R(T)$, Dimension

8m * Theorems (2)

16m * LT, $N(T)$, $R(T)$, Rank (T), Nullity (T),
* RND Theorem, 1-1 and onto (5)

UNIT IV Diagonalization.

Matrix of Linear Transformation:

Matrix of Linear Transformation (1)
Relative to the pair of bases B_v & B_w
is denoted by A or $[T]_{B_v}^{B_w}$ formed by
placing the matrix of each basis if
vectors of V column wise (i.e.)

(i.e.) $M[V_j]$ occupies j th column of $A = [T]_{B_v}^{B_w}$

$$(i.e.) A = [T]_{B_v}^{B_w} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

$$= [a_{ij}]_{m \times n}$$

where $T[V_j] = \sum_{i=1}^m a_{ij} w_i$
where $j = 1, 2, \dots, n$.

where T is a LT from V into itself.

Note: 1) If T is a LT from V into itself.
Then it is called Linear operator.

$$(i.e.) T: V \rightarrow V \text{ is a linear operator.}$$

then,

$$M[T] = A = [T]_{B_v}^{B_w} = [a_{ij}]$$

where $i = 1, 2, \dots, m$.

Standard Basis:

$\mathbb{R}^1$:

$$\mathbb{R}^1 = \{(1, 0), (0, 1)\}$$

$$\mathbb{R}^3 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$\mathbb{R}^n = \{(1, 0, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, 0, \dots, 1)\}$$

Polynomial:

$$\mathbb{P}_1(\mathbb{R}) = \{1, x\}$$

$$\mathbb{P}_2(\mathbb{R}) = \{1, x, x^2\}$$

$$\mathbb{P}_3(\mathbb{R}) = \{1, x, x^2, x^3\}$$

$$\mathbb{P}_n(\mathbb{R}) = \{1, x, x^2, \dots, x^n\}$$

Matrix:

$$M_{2 \times 2}(\mathbb{R}) = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

1) i) Let α, β are the standard basis of $\mathbb{R}^2$ & $\mathbb{R}^3$ respectively

a) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a L.T. defined by

$$T(a_1, a_2) = (2a_1 - a_2, 3a_1 + 4a_2, a_1) \text{ find } [T]$$

b) If $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is a L.T. defined by

$$T(a_1, a_2, a_3) = (a_1 - a_2, a_2) \text{ find matrix of } T \text{ relative to the standard basis of } \mathbb{R}^3 \text{ & } \mathbb{R}^2 \text{ respectively.}$$

Solution:

a) Given: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

defined by $T(a_1, a_2) = (2a_1 - a_2, 3a_1 + 4a_2, a_1)$

$$\alpha = \{ \underset{\alpha_1}{(1, 0)}, \underset{\alpha_2}{(0, 1)} \} \in \mathbb{R}^2$$

$$\beta = \{ \underset{\beta_1}{(0, 0, 0)}, \underset{\beta_2}{(0, 1, 0)}, \underset{\beta_3}{(0, 0, 1)} \} \in \mathbb{R}^3$$

$$T(v_1) = T(1, 0) = (2(1) - 0, 3(1) + 4(0), 1) = (2, 3, 1)$$

$$= 2(1, 0, 0) + 3(0, 1, 0) + 1(0, 0, 1) \\ = 2w_1 + 3w_2 + 1w_3$$

$$M[v_1] = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \rightarrow \textcircled{1}$$

$$T(v_2) = T(0, 1) = (2(0) - 1, 3(0) + 4(1), 0)$$

$$= (-1, 4, 0)$$

$$= -1(1, 0, 0) + 4(0, 1, 0) + 0(0, 0, 1)$$

$$= -1w_1 + 4w_2 + 0w_3$$

$$M[v_2] = \begin{bmatrix} -1 \\ 4 \\ 0 \end{bmatrix} \rightarrow \textcircled{2}$$

from $\textcircled{1}$ & $\textcircled{2}$, we get

$$M[T] = A = [T]_{\alpha}^{\beta} = \begin{bmatrix} 2 & -1 \\ 3 & 4 \\ 1 & 0 \end{bmatrix}_{3 \times 2}$$

b) Given: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

defined by $T(a_1, a_2, a_3) = (a_1 - a_2, a_3)$

$$\beta = \{ \underset{\beta_1}{(1, 0, 0)}, \underset{\beta_2}{(0, 1, 0)}, \underset{\beta_3}{(0, 0, 1)} \} \in \mathbb{R}^3$$

$$\alpha = \{ \underset{\alpha_1}{(1, 0)}, \underset{\alpha_2}{(0, 1)} \} \in \mathbb{R}^2$$

$$T(v_1) = T(1, 0, 0) = (1 - 0, 0) = (1, 0)$$

$$= 1(1, 0) + 0(0, 1)$$

$$M(v_1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \textcircled{1}$$

$$T(v_2) = T(0, 1, 0) = (0 - 1, 0) = (-1, 0)$$

$$= -1(1, 0) + 0(0, 1)$$

$$M(v_2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \rightarrow \textcircled{2}$$

$$T(v_3) = T(0, 0, 1) = (0 - 0, 1) = (0, 1)$$

$$= 0(1, 0) + 1(0, 1)$$

$$= 0w_1 + 1w_2$$

$$M(v_3) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \textcircled{3}$$

From ①, ② & ③, we get

$$M[T] = A = [T]_{\beta}^{\alpha} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

14/8/25 (ii) Let $\alpha = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$

$\beta = \{f_1, x, x^2\}$ are the standard basis of $\mathbb{R}$ & $\mathbb{P}_2(\mathbb{R})$ respectively.

$M_{\alpha\alpha}$

$\mathbb{R} \rightarrow M_{\alpha\alpha}(\mathbb{R})$ via a linear

a) Define $T: M_{\alpha\alpha}(\mathbb{R}) \rightarrow M_{\alpha\alpha}(\mathbb{R})$ as a linear transformation by $T(A) = A^t$, find $[T]_{\alpha}$

b) Let $T: \mathbb{P}_2(\mathbb{R}) \rightarrow M_{\alpha\alpha}(\mathbb{R})$ be a linear transformation defined by $T(f(x)) = \begin{bmatrix} f(0) & f'(0) \\ 0 & f(0) \end{bmatrix}$ compare $[T]_{\beta}$

Solution:

1a) Given: $T: M_{\alpha\alpha}(\mathbb{R}) \rightarrow M_{\alpha\alpha}(\mathbb{R})$ by $T(A) = A^t$

$$\alpha = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

$$T(v_1) = T \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= 1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$= 1w_1 + 0w_2 + 0w_3$$

$$M(v_1) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow 0$$

$$T(v_2) = T \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= 0w_1 + 0w_2 + 1w_3 + 0w_4$$

$$M(v_2) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \rightarrow ③$$

$$T(v_3) = T \left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= 0w_1 + 1w_2 + 0w_3 + 0w_4$$

$$M(v_3) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow ⑤$$

$$T(v_4) = T \left(\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= 0w_1 + 0w_2 + 0w_3 + 1w_4$$

$$M(v_4) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow ④$$

From ①, ②, ③ & ④, we get

$$M[T] = A = [T]_{\alpha} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4}$$

1b) Given: $T: \mathbb{P}_2(\mathbb{R}) \rightarrow M_{\alpha\alpha}(\mathbb{R})$ by

$$T(f(x)) = \begin{bmatrix} f'(0) & 2f(1) \\ 0 & f''(0) \end{bmatrix}$$

$$\alpha = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$$

$$\beta = \{1, x, x^2\}$$

$$\begin{aligned} T(v_1) &= T(1) = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{matrix} f'(x)=1 \\ f'(x)=0 \\ f''(x)=0 \end{matrix} \\ &= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ &= 0w_1 + 2w_2 + 0w_3 + 0w_4 \end{aligned}$$

$$M(v_1) = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} \rightarrow \textcircled{1}$$

$$\begin{aligned} T(v_2) &= T(x) = \begin{bmatrix} 1 & 2x(1) \\ 0 & 0 \end{bmatrix} \begin{matrix} f'(x)=x \\ f'(x)=1 \\ f''(x)=1 \end{matrix} \\ &= \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \\ &= 1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ &= 1w_1 + 2w_2 + 0w_3 + 0w_4 \end{aligned}$$

$$M(v_2) = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} \rightarrow \textcircled{2}$$

$$\begin{aligned} T(v_3) &= T(x^2) = \begin{bmatrix} 0 & 2x(1) \\ 0 & 2 \end{bmatrix} \begin{matrix} f'(x)=2x \\ f'(x)=2 \\ f''(x)=2 \end{matrix} \\ &= \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \\ &= 0 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + 0 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ &= 0w_1 + 2w_2 + 0w_3 + 2w_4 \end{aligned}$$

$$M(v_3) = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \rightarrow \textcircled{3}$$

from (1), (2) & (3), we get

$$M[T] = A = [T]_{\beta}^{\alpha} = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}_{4 \times 3}$$

ii) Let $\alpha = \{1, x, x^2\} \in \mathcal{B}_2(\mathbb{R})$, $\beta = \{1, x, x^2, x^3\} \in \mathcal{B}_3(\mathbb{R})$ are the standard bases of $M_{2 \times 2}(\mathbb{R})$, $\mathcal{B}_2(\mathbb{R})$ respectively.

Define $T: \mathcal{B}_2(\mathbb{R}) \rightarrow \mathcal{B}_3(\mathbb{R})$ as a linear transformation by $T[f(x)] = xf(x) + f'(x)$, and T relative to the standard bases of $\mathcal{B}_2(\mathbb{R})$, $\mathcal{B}_3(\mathbb{R})$ respectively.

Let $T: M_{2 \times 2}(\mathbb{R}) \rightarrow \mathcal{B}_3(\mathbb{R})$ be a linear transformation defined by

$T \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = (actb) + (adt) + bct$. compute matrix of T to the standard bases of $M_{2 \times 2}(\mathbb{R})$ & $\mathcal{B}_3(\mathbb{R})$ respectively.

Solution:

a) Define $T: \mathcal{B}_2(\mathbb{R}) \rightarrow \mathcal{B}_3(\mathbb{R})$ by

$$T[f(x)] = xf(x) + f'(x)$$

$$\beta = \{1, x, x^2\}, \gamma = \{1, x, x^2, x^3\}$$

$$T[v_1] = T(1) = x(1) + 0 = x$$

$$= 0.1 + 1.x + 0.x^2 + 0.x^3 \\ = 0w_1 + 1w_2 + 0w_3 + 0w_4$$

$$M[V_1] = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \rightarrow \textcircled{1}$$

$$T(V_2) = T(x) = x(x) + 1 = x^2 + 1$$

$$= 1.1 + 0.x + 1.x^2 + 0.x^3 \\ = 1w_1 + 0w_2 + 1w_3 + 0w_4$$

$$M(V_2) = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \textcircled{2}$$

$$T(V_3) = T(x^2) = x.x^2 + 2x = x^3 + 2x \\ = 0.1 + 2x + 0.x^2 + 1.x^3 \\ = 0w_1 + 2w_2 + 0w_3 + 1w_4$$

$$M[V_3] = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \rightarrow \textcircled{3}$$

$$\text{Hence } M[T] = [T]_{\beta}^{\delta} = A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{4 \times 3}$$

(using $\textcircled{1}$, $\textcircled{2}$ & $\textcircled{3}$)

$$\text{Defn: } T: M_{2 \times 2}(R) \longrightarrow M_2(R) \text{ by}$$

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b)(c+d)x + b x^2$$

$$\alpha = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$$\beta = \{1, x, x^2\}$$

$$T(V_1) = T\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right) = (1+0) + [2(0)]x + 0x^2 \\ = 1 + 0 + 0 = 1$$

$$= 1.1 + 0.x + 0.x^2 \\ = 1w_1 + 0w_2 + 0w_3$$

$$M(V_1) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \textcircled{1}$$

$$T(V_2) = T\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\right) = (0+1) + [2(0)]x + 1.x^2 \\ = 1 + 0 + x^2 = 1 + x^2$$

$$= 1.1 + 0.x + 1.x^2 \\ = 1w_1 + 0w_2 + 1w_3$$

$$M(V_2) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \rightarrow \textcircled{2}$$

$$T(V_3) = T\left(\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\right) = (0+0) + [2(0)]x + 0.x^2 \\ = 0$$

$$= 0.1 + 0.x + 0.x^2 \\ = 0w_1 + 0w_2 + 0w_3$$

$$M(V_3) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \textcircled{3}$$

$$T(V_4) = T\left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 \end{bmatrix}\right) = (0+0) + [2(1)]x + 0x^2$$

$$= 0 + 2x + 0 = 2x$$

$$= 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2$$

$$= 0\omega_0 + 2\omega_2 + 0\omega_3$$

$$M(V_4) = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \rightarrow \textcircled{A}$$

From $\textcircled{D}$, $\textcircled{E}$, $\textcircled{F}$ & $\textcircled{A}$, we get

$$\text{Hence, } M[T] = [T]_{\alpha}^{\beta} = A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Find the matrix of $L.T$ on $\mathbb{R}^3$ defined as $T(a, b, c) = (2b+c, a-4b, 3a)$ with respect to ordered basis $\beta = \{v_1, v_2, v_3\}$

Solution: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$\text{Given: } T(a, b, c) = (2b+c, a-4b, 3a)$$

$$\text{by } T(a, b, c) = (2b+c, a-4b, 3a)$$

$$\text{ordered Basis } \beta = \{v_1, v_2, v_3\}$$

$$T(v_1) = T(1, 1, 1) = (2(1)+1, 1-4(1), 3(1)) = (3, -3, 3)$$

$$(3, -3, 3) = x(1, 1, 1) + y(1, 1, 0) + z(1, 0, 0)$$

$$3 = x + y + z \rightarrow \textcircled{1}$$

$$-3 = x + y + 0z \rightarrow \textcircled{2}$$

$$3 = x + 0y + 0z \rightarrow \textcircled{3}$$

$$3 = x + 0y + 0z \rightarrow \textcircled{3}$$

On solving $\textcircled{1}$, $\textcircled{2}$ & $\textcircled{3}$, we get

$$\boxed{x=3, y=-6, z=b}$$

$$\therefore (3, -3, 3) = 3(1, 1, 1) - 6(1, 1, 0) + b(1, 0, 0)$$

$$T(v_1) = 3\omega_0 - 6\omega_2 + b\omega_3$$

$$\therefore M(v_1) = \begin{bmatrix} 3 \\ -6 \\ b \end{bmatrix} \rightarrow \textcircled{A}$$

$$T(v_2) = T(1, 1, 0) = (2(1)+0, 1-4(1), 3(1, 0)) = (2, -3, 3)$$

$$(2, -3, 3) = x_1(1, 1, 1) + y_1(1, 1, 0) + z_1(1, 0, 0)$$

$$x_1 + y_1 + z_1 = 2 \rightarrow \textcircled{A}$$

$$x_1 + y_1 + 0x_2 = -3 \rightarrow \textcircled{5}$$

$$x_1 + 0y_1 + 0x_2 = 3 \rightarrow \textcircled{6}$$

On solving $\textcircled{4}$, $\textcircled{5}$ & $\textcircled{6}$, we get

$$\boxed{x_1 = 3}, \quad \boxed{y_1 = -6}, \quad \boxed{z_1 = 5}$$

$$\therefore (2, -3, 3) = 3(1, 1, 1) - 6(1, 1, 0) + 5(1, 0, 0)$$

$$T(v_2) = 3w_1 - 6w_2 + 5w_3$$

$$\therefore M(v_2) = \begin{bmatrix} 3 \\ -6 \\ 5 \end{bmatrix} \rightarrow \textcircled{8}$$

$$T(v_3) = T(1, 0, 0) = (2, 0, 0) + 0, 1 - 4(0, 3(1))$$

$$= (0, 1, 3)$$

$$(0, 1, 3) = x_2(1, 1, 1) + y_2(1, 1, 0) + z_2(1, 0, 0)$$

$$\Rightarrow x_2 + y_2 + z_2 = 0 \rightarrow \textcircled{7}$$

$$x_2 + y_2 + 0z_2 = 1 \rightarrow \textcircled{8}$$

$$x_2 + 0y_2 + 0z_2 = 3 \rightarrow \textcircled{9}$$

On solving $\textcircled{7}$, $\textcircled{8}$ & $\textcircled{9}$, we get

$$\boxed{x_2 = 3}, \quad \boxed{y_2 = -2}, \quad \boxed{z_2 = -1}$$

$$\therefore (0, 1, 3) = 3(1, 1, 1) - 2(1, 1, 0) - 1(1, 0, 0)$$

$$T(v_3) = 3w_1 - 2w_2 - 1w_3$$

$$M(v_3) = \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix} \rightarrow \textcircled{9}$$

From $\textcircled{7}$, $\textcircled{8}$ & $\textcircled{9}$, we get

$$M(T) = [T]_{\beta} = A = \begin{bmatrix} 3 & 3 & 3 \\ -6 & -6 & -2 \\ 5 & 5 & -1 \end{bmatrix}_{3 \times 3}$$

2.)

a) i) Find the matrix of $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined as $T(a_1, a_2) = (a_1 - a_2, a_1 + 2a_1 + a_2)$ with standard basis of $\mathbb{R}^2$ as $\{(1, 0), (0, 1)\}$ & ordered basis of $\mathbb{R}^3$ as $\{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$.

Solution:

$$\text{Given: } T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

$$\text{Given: } T: (a_1, a_2) = (a_1 - a_2, a_1 + a_2)$$

$$\text{by } T(a_1, a_2) = \{(1, 0), (0, 1)\}$$

$$\text{standard basis} = \{(1, 0), (0, 1)\}$$

$$\text{ordered basis } v = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$$

$$T(v_1) = T(1, 0) = (1, 1, 2) \in \mathbb{R}^3$$

Now express $(1, 1, 2)$ as a linear combination of vectors (v) such that,

$$(1, 1, 2) = x(1, 1, 0) + y(0, 1, 1) + z(2, 2, 3)$$

$$= (x + 0y + 2z, x + y + 2z, 0x + y + 3z)$$

$$x + 0y + 2z = 1 \rightarrow \textcircled{1}$$

$$x + y + 2z = 1 \rightarrow \textcircled{2}$$

$$0x + y + 3z = 2 \rightarrow \textcircled{3}$$

On solving $\textcircled{1}$, $\textcircled{2}$ & $\textcircled{3}$, we get,

$$x = -0.3333 = -1/3$$

$$y = 0$$

$$z = 0.6667 = 2/3$$

$$T(v_1) = -1/3(1, 1, 0) + 0(0, 1, 1) + 2/3(2, 2, 3)$$

$$= -\frac{1}{8}w_1 + 0w_2 + \frac{7}{8}w_3$$

$$M(V_1) = \begin{bmatrix} -\frac{1}{8} \\ 0 \\ \frac{7}{8} \end{bmatrix} \rightarrow \textcircled{A}$$

$$\text{Now } T(V_2) = T(0, 1) = (-1, 0, 1)$$

Express $0(-1, 0, 1)$ as a linear combination of vectors (V) such that

$$(-1, 0, 1) = x_1(1, 1, 0) + x_2(0, 1, 1) + x_3(2, 2, 3)$$

$$-1 = x_1 + 0x_2 + 2x_3 \rightarrow \textcircled{4}$$

$$0 = x_1 + x_2 + 2x_3 \rightarrow \textcircled{5}$$

$$1 = 0x_1 + x_2 + 3x_3 \rightarrow \textcircled{6}$$

On solving $\textcircled{4}$, $\textcircled{5}$ & $\textcircled{6}$, we get,

$$\boxed{x_1 = -1}, \boxed{x_2 = 1}, \boxed{x_3 = 0}$$

$$(-1, 0, 1) = -1(1, 1, 0) + 1(0, 1, 1) + 0(2, 2, 3)$$

$$T(V_2) = -1w_1 + 1w_2 + 0w_3$$

$$M(V_2) = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \rightarrow \textcircled{B}$$

From $\textcircled{A}$ & $\textcircled{B}$ we get,

$$M(T) = [T]_{\beta}^{\alpha} = A = \begin{bmatrix} -\frac{1}{8} & -1 \\ 0 & 1 \\ \frac{7}{8} & 0 \end{bmatrix}_{3 \times 2}$$

iii)

Find the matrix of $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as $T(a_1, a_2, a_3) = (2a_1 + 3a_2 - a_3, a_1 + a_2)$. If ordered basis $\alpha = \{(1, 2), (0, 3)\}$, $\beta = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$ of $\mathbb{R}^3, \mathbb{R}^2$ respectively, then compute $[T]_{\beta}^{\alpha}$.

Solution: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$\text{given: } T(a_1, a_2, a_3) = (2a_1 + 3a_2 - a_3, a_1 + a_2)$$

Given ordered basis of $\mathbb{R}^3$ is

$$\beta = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$$

ordered basis of $\mathbb{R}^2$ is

$$\alpha = \{(1, 2), (2, 3)\}$$

$$T(V_1) = T(1, 1, 0) = (2(1) + 3(1) - 0, 1 + 1) = (5, 2)$$

Now, we express $(5, 2)$ as a linear combination of vectors α such that

$$(5, 2) = x_1(1, 2) + x_2(2, 3)$$

$$\Rightarrow 5 = x_1 + 2x_2 \rightarrow \textcircled{1}$$

$$2 = x_1 + 3x_2 \rightarrow \textcircled{2}$$

On solving $\textcircled{1}$ & $\textcircled{2}$, we get,

$$\boxed{x_1 = -11}, \boxed{x_2 = 8}$$

$$\therefore (5, 2) = -11(1, 2) + 8(2, 3)$$

$$M(V_1) = [8] \rightarrow \textcircled{1}$$

$$T(V_2) = T(0, 1, 1) = (2(0) + 3(1) - 1, 0 + 1) = (2, 1)$$

Also we express $(2,1)$ as a linear combination of vectors α such that

$$(2,1) = x_3(1,2) + x_4(2,3)$$

$$\Rightarrow 2 = x_3 + 2x_4 \rightarrow (3)$$

$$1 = 2x_3 + 3x_4 \rightarrow (4)$$

On solving (3) & (4), we get,

$$\boxed{x_3 = -4}; \boxed{x_4 = 3}$$

$$(2,1) = -4(1,2) + 3(2,3)$$

$$M(v_2) = \begin{bmatrix} -4 \\ 3 \end{bmatrix} \rightarrow (5)$$

$$T(v_3) = T(2,2,3) = (2(2) + 3(2) - 3, 2+2) = (7,4)$$

Now, we express $(7,4)$ as a linear combination of vector α such that

$$(7,4) = x_5(1,2) + x_6(2,3)$$

$$\Rightarrow 7 = x_5 + 2x_6 \rightarrow (6)$$

$$4 = 2x_5 + 3x_6 \rightarrow (7)$$

On solving (6) & (7), we get

$$\boxed{x_5 = -13}; \boxed{x_6 = 10}$$

$$\therefore (7,4) = -13(1,2) + 10(2,3)$$

$$M(v_3) = \begin{bmatrix} -13 \\ 10 \end{bmatrix} \rightarrow (8)$$

From (3), (4), (6), (7), we get

$$M[T] = [T]_{\beta}^{\alpha} = A = \begin{bmatrix} -11 & -4 & -13 \\ 8 & 3 & 10 \end{bmatrix}_{2 \times 3}$$

If T is a linear on $V = \mathbb{R}^2$ defined by $T(a,b) = (-2a+3b, -10a+9b)$ find the values of T and an ordered basis β for V such that $[T]_{\beta}$ is a diagonal matrix.

Solution:

$$\text{Given: } T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\text{by } T(a,b) = (-2a+3b, -10a+9b)$$

$$\beta = \{(1,0), (0,1)\}$$

$$T(v_1) = T(1,0) = (-2(1) + 3(0), -10(1) + 9(0)) = (-2, -10)$$

$$\text{Now, } (-2, -10) = (-2)(1,0) + (-10)(0,1)$$

$$T(v_1) = \begin{bmatrix} -2 \\ -10 \end{bmatrix} \rightarrow (1)$$

$$\therefore M(v_1) = \begin{bmatrix} -2 \\ -10 \end{bmatrix} \rightarrow (1)$$

$$T(v_2) = T(0,1) = (-2(0) + 3(1), -10(0) + 9(1)) = (3, 9)$$

$$\text{Also, } (3,9) = 3(1,0) + 9(0,1)$$

$$T(v_2) = 3v_1 + 9v_2$$

$$\therefore M(v_2) = \begin{bmatrix} 3 \\ 9 \end{bmatrix} \rightarrow (2)$$

From (1) & (2), we get,

$$M[T] = [T]_{\beta}^{\beta} = A = \begin{bmatrix} -2 & 3 \\ -10 & 9 \end{bmatrix}_{2 \times 2}$$

which is not a diagonal matrix.

3) ii) If T is linear on $V = \mathbb{R}^3$ defined by

$$T(a, b, c) = (7a - 4b + 10c, 4a - 3b + 8c, -2a + b - 2c)$$

Find the eigen values of T and an ordered basis β for V such that $[T]_{\beta}$ is a diagonal matrix.

Solution:

$$\text{On } V: T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$\text{by } T(a, b, c) = (7a - 4b + 10c, 4a - 3b + 8c, -2a + b - 2c)$$

$$\beta = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$T(v_1) = T(1, 0, 0) = (7(1) - 4(0) + 10(0), 4(1) - 3(0) + 8(0), -2(1) + 1(0) - 2(0))$$

$$= (7, 4, -2)$$

$$= 7(1, 0, 0) + 4(0, 1, 0) - 2(0, 0, 1)$$

$$= 7v_1 + 4v_2 - 2v_3$$

$$M(V) = \begin{bmatrix} 7 & & \\ & 4 & \\ & & -2 \end{bmatrix} \rightarrow \text{Diagonal}$$

$$T(v_2) = T(0, 1, 0) = (7(0) - 4(1) + 10(0), 4(0) - 3(1) + 8(0), -2(0) + 1(1) - 2(0))$$

$$= (-4, -3, 1)$$

$$= -4(1, 0, 0) - 3(0, 1, 0) + 1(0, 0, 1)$$

$$= -4v_1 - 3v_2 + 1v_3$$

$$\therefore M(V) = \begin{bmatrix} 7 & & \\ & 4 & \\ & & -2 \end{bmatrix} \rightarrow \text{Diagonal}$$

$$T(v_3) = T(0, 0, 1) = (7(0) - 4(0) + 10(1), 4(0) - 3(0) + 8(1), -2(0) + 0(0) - 2(1))$$

$$= (10, 8, -2)$$

$$= 10(1, 0, 0) + 8(0, 1, 0) - 2(0, 0, 1)$$

$$= 10v_1 + 8v_2 - 2v_3$$

$$M(V) = \begin{bmatrix} 10 & & \\ & 8 & \\ & & -2 \end{bmatrix} \rightarrow \text{Diagonal}$$

From Q. 3) a) we get

$$M[T] = [T]_{\beta} = A = \begin{bmatrix} 7 & -4 & 10 \\ 4 & -3 & 8 \\ -2 & 1 & -2 \end{bmatrix}_{3 \times 3}$$

To find the eigen values & eigen vector

To characteristic equation is

$$\lambda^3 - 5\lambda^2 + 8\lambda - 33 = 0$$

$$5\lambda^3 - 5\lambda^2 - 2 = 2$$

$$5\lambda = \begin{vmatrix} -3 & 8 \\ 1 & -2 \end{vmatrix} + \begin{vmatrix} 7 & 10 \\ 4 & -3 \end{vmatrix} + \begin{vmatrix} 7 & -4 \\ -2 & 1 \end{vmatrix}$$

$$= (6-8) + (-14+20) + (-2+4)$$

$$= -2+6-5$$

$$5\lambda = -1$$

$$S_3 = |A| = \begin{vmatrix} 7 & -4 & 10 \\ 4 & -3 & 8 \\ -2 & 1 & -2 \end{vmatrix} = 7(6-8) + 4(-8+16) + 10(4-6)$$

$$= 7(-2) + 4(8) + 10(-2)$$

$$= -14 + 32 - 20$$

$$= -2$$

$\therefore$ The characteristic equation is

$$\lambda^3 - 2\lambda^2 - 1\lambda + 2 = 0 \rightarrow \textcircled{1}$$

On solving $\textcircled{1}$, we get

The Eigen values are

$$\lambda = 2, -1, 1$$

To find Eigen vectors,

$$\text{Let } [A - \lambda I] X = 0$$

$$\begin{bmatrix} 7-\lambda & -4 & 10 \\ 4 & -3-\lambda & 8 \\ -2 & 1 & -2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\rightarrow (7-\lambda)x_1 - 4x_2 + 10x_3 = 0 \rightarrow \textcircled{4}$$

$$4x_1 + (-3-\lambda)x_2 + 8x_3 = 0 \rightarrow \textcircled{5}$$

$$-2x_1 + 1x_2 + (-2-\lambda)x_3 = 0 \rightarrow \textcircled{6}$$

Case (i): put $\lambda = 2$ in $\textcircled{4}$, $\textcircled{5}$ & $\textcircled{6}$, we get

$$5x_1 - 4x_2 + 10x_3 = 0 \rightarrow \textcircled{7}$$

$$4x_1 - 5x_2 + 8x_3 = 0 \rightarrow \textcircled{8}$$

$$-2x_1 + x_2 - 4x_3 = 0 \rightarrow \textcircled{9}$$

Here all three equations are different using cross-multiplication rule.

By $\textcircled{7}$ & $\textcircled{8}$,

$$\begin{array}{ccc} -4 & 10 & -4 \\ \swarrow & \searrow & \swarrow \\ 8 & 4 & -5 \end{array}$$

$$\frac{x_1}{-32+50} = \frac{x_2}{40-40} = \frac{x_3}{-25+16}$$

$$\frac{x_1}{18} = \frac{x_2}{0} = \frac{x_3}{-9} = k_1 \text{ (say)}$$

$$X_1 = \begin{bmatrix} 18k_1 \\ 0 \\ -9k_1 \end{bmatrix}$$

$$\text{put } k_1 = \frac{1}{9}$$

$X_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$ be the Eigen vector corresponding to $\lambda = 2$.

Case (ii): put $\lambda = -1$ in $\textcircled{4}$, $\textcircled{5}$ & $\textcircled{6}$, we get

$$8x_1 - 4x_2 + 10x_3 = 0 \rightarrow \textcircled{10}$$

$$4x_1 - 2x_2 + 8x_3 = 0 \rightarrow \textcircled{11}$$

$$-2x_1 + x_2 - x_3 = 0 \rightarrow \textcircled{12}$$

Here all three equations are different.

So, consider $\textcircled{10}$ & $\textcircled{12}$

$$\begin{array}{ccc} -2 & 8 & -2 \\ \swarrow & \searrow & \swarrow \\ -1 & -2 & 1 \end{array}$$

Using cross-multiplication rule,

$$\frac{x_1}{2-8} = \frac{x_2}{-16+4} = \frac{x_3}{4-4} = k_2 (\text{say})$$

$$\frac{x_1}{-6} = \frac{x_2}{-12} = \frac{x_3}{0} = k_2$$

$$x_2 = \begin{bmatrix} -6k_2 \\ -12k_2 \\ 0k_2 \end{bmatrix}$$

put $k_2 = -\frac{1}{6}$

$x_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ be the Eigen vectors corresponding to $\lambda = -1$

case (iii):

put $\lambda = 1$ in (8), (9) & (10), we get,

$$6x_1 - 4x_2 + 10x_3 = 0 \rightarrow (8)$$

$$4x_1 - 4x_2 + 8x_3 = 0 \rightarrow (14)$$

$$-2x_1 - 12x_2 - 3x_3 = 0 \rightarrow (15)$$

Here all three equations are different.

So, we choose (8) & (14),

$$\begin{array}{ccc} -4 & 10 & 6 \\ & \searrow & \nearrow \\ & 8 & 4 \end{array} \quad \begin{array}{ccc} 4 & -4 & 8 \\ & \searrow & \nearrow \\ & 4 & -4 \end{array}$$

using cross-multiplication rule,

$$\frac{x_1}{-32+40} = \frac{x_2}{40-48} = \frac{x_3}{-24+16} = k_3 (\text{say})$$

$$x_3 = \begin{bmatrix} 8k_3 \\ -8k_3 \\ -8k_3 \end{bmatrix}$$

put $k_3 = \frac{1}{8}$

$x_3 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$ be the Eigen vectors corresponding to $\lambda = 1$

Result:

$$A = [T]_B = \begin{bmatrix} 7 & -4 & 10 \\ 4 & -3 & 8 \\ -2 & 1 & -2 \end{bmatrix}$$

characteristic equation: $\lambda^3 - 2\lambda^2 - \lambda + 2 = 0$

Eigen values

Eigen vectors

$$\lambda_1 = 2$$

$$x_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

$$\lambda_2 = -1$$

$$x_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$\lambda_3 = 1$$

$$x_3 = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

since, $B = \{x_1, x_2, x_3\}$ is a required ordered basis in which $[T]_B$ is a diagonal matrix.

3) iii)

If T is a linear on $V = \mathbb{R}^3$ defined by $T(a, b, c) = (-4a + 3b - 6c, 6a - 7b + 12c, 6a - 6b + 11c)$. Find the eigen values of T & an ordered basis β for V such that $[T]_\beta$ is a diagonal matrix.

Solution:

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by

$$T(a, b, c) = (-4a + 3b - 6c, 6a - 7b + 12c, 6a - 6b + 11c)$$

$$\beta = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$T(v_1) = T(1, 0, 0) = \{(-4(1) + 3(0) - 6(0), 6(1) - 7(0) + 12(0), 6(1) - 6(0) + 11(0))\}$$

$$= (-4, 6, 6)$$

$$= -4(1, 0, 0) + 6(0, 1, 0) + 6(0, 0, 1)$$

$$= -4w_1 + 6w_2 + 6w_3$$

$$M(v_1) = \begin{bmatrix} -4 \\ 6 \\ 6 \end{bmatrix} \rightarrow \textcircled{1}$$

$$T(v_2) = T(0, 1, 0) = \{-4(0) + 3(1) - 6(0), 6(0) - 7(1) + 12(0), 6(0) - 6(1) + 11(0)\}$$

$$= (3, -7, -6)$$

$$= 3(1, 0, 0) - 7(0, 1, 0) - 6(0, 0, 1)$$

$$= 3w_1 - 7w_2 - 6w_3$$

$$M(v_2) = \begin{bmatrix} 3 \\ -7 \\ -6 \end{bmatrix} \rightarrow \textcircled{2}$$

$$T(v_3) = T(0, 0, 1) = \{-4(0) + 3(0) - 6(1), 6(0) - 7(0) + 12(1), 6(0) - 6(0) + 11(1)\}$$

$$= (-6, 12, 11)$$

$$= -6(1, 0, 0) + 12(0, 1, 0) + 11(0, 0, 1)$$

$$= -6w_1 + 12w_2 + 11w_3$$

$$M(v_3) = \begin{bmatrix} -6 \\ 12 \\ 11 \end{bmatrix} \rightarrow \textcircled{3}$$

From $\textcircled{1}$, $\textcircled{2}$ & $\textcircled{3}$, we get

$$M[T] = [T]_\beta = A = \begin{bmatrix} -4 & 3 & -6 \\ 6 & -7 & 12 \\ 6 & -6 & 11 \end{bmatrix}$$

The characteristic equation is

$$\lambda^3 - 8\lambda^2 + 8\lambda - 83 = 0$$

$$\lambda_1 = -4 - 7 + 11 = -11 + 11 = 0$$

$$\lambda_2 = \begin{vmatrix} -7 & 12 \\ -6 & 11 \end{vmatrix} + \begin{vmatrix} -4 & -6 \\ 6 & -7 \end{vmatrix} + \begin{vmatrix} -4 & 3 \\ 6 & -7 \end{vmatrix}$$

$$= (-77 + 72) + (-44 + 36) + (28 - 18) = -5 - 8 + 10$$

$$= -13 + 10 = -3$$

$$\lambda_3 = -3$$

$$\lambda_3 = \begin{vmatrix} -4 & 3 \\ 6 & -7 \end{vmatrix} + \begin{vmatrix} -7 & 12 \\ -6 & 11 \end{vmatrix} + \begin{vmatrix} -4 & -6 \\ 6 & -7 \end{vmatrix}$$

$$= -4(-77 + 72) - 3(66 - 72) - 6(-36 + 42)$$

$$= -4(-5) - 3(-6) - 6(6)$$

$$= 20 + 18 - 36$$

$$= 38 - 36$$

$$\lambda_3 = 2$$

∴ The characteristic equation is $\lambda^3 - 0\lambda^2 - 3\lambda - 12 = 0 \rightarrow \textcircled{1}$

On solving $\textcircled{1}$, we get the Eigen values are

$$\lambda = 2, -1,$$

To find Eigen vectors

Let $[A - \lambda I]x = 0$

$$\begin{bmatrix} -4-\lambda & 3 & -6 \\ 6 & -7-\lambda & 12 \\ 6 & 6 & 11-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(-4-\lambda)x_1 + 3x_2 - 6x_3 = 0 \rightarrow \textcircled{4}$$

$$6x_1 + (-7-\lambda)x_2 + 12x_3 = 0 \rightarrow \textcircled{5}$$

$$6x_1 - 6x_2 + (11-\lambda)x_3 = 0 \rightarrow \textcircled{6}$$

Case (i):

put $\lambda = 2$ in $\textcircled{4}$, $\textcircled{5}$ & $\textcircled{6}$, we get

$$-6x_1 + 3x_2 - 6x_3 = 0 \rightarrow \textcircled{7}$$

$$6x_1 - 9x_2 + 12x_3 = 0 \rightarrow \textcircled{8}$$

$$6x_1 - 6x_2 + 9x_3 = 0 \rightarrow \textcircled{9}$$

Here all three equations are different using cross-multiplication rule,

By $\textcircled{7}$ & $\textcircled{8}$

$$\begin{array}{ccc} 3 & -6 & -6 \\ \swarrow & \searrow & \swarrow \\ -9 & 12 & 6 \end{array} \begin{array}{ccc} -6 & -6 & 3 \\ \swarrow & \searrow & \swarrow \\ 12 & 6 & -9 \end{array}$$

$$\frac{x_1}{86-54} = \frac{x_2}{-36+12} = \frac{x_3}{54-18}$$

$$\frac{x_1}{-18} = \frac{x_2}{36} = \frac{x_3}{36} = k, \text{ (say)}$$

$$x_1 = \begin{bmatrix} -18k \\ 36k \\ 36k \end{bmatrix}$$

$$\text{put } k = \frac{1}{18}$$

$$x_1 = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \text{ be the Eigen vector corresponding to } \lambda = 2$$

Case (ii):

put $\lambda = -1$ in $\textcircled{4}$, $\textcircled{5}$ & $\textcircled{6}$, we get

$$-3x_1 + 3x_2 - 6x_3 = 0 \rightarrow \textcircled{10}$$

$$6x_1 - 6x_2 + 12x_3 = 0 \rightarrow \textcircled{11}$$

$$6x_1 - 6x_2 + 12x_3 = 0 \rightarrow \textcircled{12}$$

31/11
 $T(a,b) = (-2a+3b, -10a+9b)$ find the values of T and an ordered basis β for V such that $[T]_\beta$ is a diagonal matrix.

Solution:

Given: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

by $T(a,b) = (-2a+3b, -10a+9b)$

$$\beta = \{(1,0), (0,1)\}$$

$$T(v_1) = T(1,0)$$

$$= (-2(1) + 3(0), -10(1) + 9(0))$$

$$= (-2, -10)$$

$$= -2(1,0) + (-10)(0,1)$$

$$= -2w_1 - 10w_2$$

$$M(v_1) = \begin{bmatrix} -2 \\ -10 \end{bmatrix} \rightarrow 0 \cdot x + (-10) \cdot y + 0 \cdot z$$

$$T(v_2) = T(0,1)$$

$$= (-2(0) + 3(1), -10(0) + 9(1))$$

$$= (3, 9)$$

$$= 3(1,0) + 9(0,1)$$

$$= 3w_1 + 9w_2$$

$$M(v_2) = \begin{bmatrix} 3 \\ 9 \end{bmatrix} \rightarrow 0 \cdot x + 3 \cdot y + 9 \cdot z$$

From ① & ②, we get,

$$M[T]_\beta = [T]_\beta = A = \begin{bmatrix} -2 & 3 \\ -10 & 9 \end{bmatrix}$$

To find characteristic equation

$$\lambda^2 - 8\lambda + 8 = 0$$

$$\lambda_1 = 4, \lambda_2 = 4$$

$$\lambda_2 = -10, 9$$

$$= -18 + 30$$

$\therefore$ The characteristic equation is

$$\lambda^2 - 7\lambda + 12 = 0$$

On solving ③, we get

The eigen values are

$$\lambda = 4, 3$$

To find Eigen vectors

$$\text{Let } [A - \lambda I]x = 0$$

$$\begin{bmatrix} -2-\lambda & 3 \\ -10 & 9-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$(-2-\lambda)x_1 + 3x_2 = 0$$

$$-10x_1 + (9-\lambda)x_2 = 0$$

Case (i): put $\lambda = 4$ in ③ & ④, we get

$$-6x_1 + 3x_2 = 0$$

$$-10x_1 + 5x_2 = 0$$

consider equation ⑤

$$-6x_1 = -3x_2$$

$$\frac{x_1}{7x_1} = \frac{x_2}{4x_2}$$

$$x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Ques (ii): put $\lambda = 3$ in (3) & (4), we get

$$-5x_1 + 3x_2 = 0 \rightarrow (1)$$

$$-10x_1 + 6x_2 = 0 \rightarrow (2)$$

consider equation (1) $\Rightarrow$ eqn (2) is satisfied

$$-5x_1 = -3x_2 \Rightarrow x_1 = \frac{3}{5}x_2$$

$$x_1 = \frac{3}{5}x_2$$

$$x_2 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Result:

$$A = [T]_B = \begin{bmatrix} -2 & 3 \\ -10 & 5 \end{bmatrix} = [T]_B [D]_B [T]_B^{-1}$$

Characteristic equation is $\lambda^2 - 7\lambda + 12 = 0$

$$\lambda^2 - 7\lambda + 12 = 0$$

Eigen values are $\lambda_1 = 4$ and $\lambda_2 = 3$

$$\lambda_1 = 4$$

$$\lambda_2 = 3$$

Since, $B = [x_1, x_2]$ is a required

ordered basis in which $[T]_B$ is a

diagonal matrix.

11/1/25

Diagonalizability:

Def for

T is a linear operator

on an n -dimensional vector space

then T is diagonalizable. If

a) the characteristic polynomial

on the form

$$p(\lambda) = (\lambda - \lambda_1)^{m_1} (\lambda - \lambda_2)^{m_2} \dots (\lambda - \lambda_k)^{m_k}$$

b) For each eigen value λ_i

$$\dim[E_{\lambda_i}] = m_i \quad i=1 \leq i \leq k$$

this condition is satisfied for an

eigenvalue with multiplicity ≥ 1 . Even if

one condition fails then T is not

diagonalizable.

$$D = Q^{-1} A Q \text{ is a diagonal matrix}$$

where Q is a matrix of

eigen vectors of A .

$$\rightarrow \begin{bmatrix} 1-\lambda & 4 \\ 8 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\rightarrow (1-\lambda)x_1 + 4x_2 = 0 \rightarrow \textcircled{3}$$

$$8x_1 + (2-\lambda)x_2 = 0 \rightarrow \textcircled{4}$$

case (i): put $\lambda = 5$ in eqn $\textcircled{3}$ & $\textcircled{4}$, we get,

$$-4x_1 + 4x_2 = 0 \rightarrow \textcircled{5}$$

$$8x_1 - 3x_2 = 0 \rightarrow \textcircled{6}$$

Here $\textcircled{5}$ & $\textcircled{6}$ are same

$$\text{So, } -4x_1 = -4x_2$$

$$\frac{x_1}{x_2} = \frac{-4x_2}{-4x_2} = 1 \text{ (say)}$$

$$\Rightarrow x_1 = \begin{bmatrix} k_1 \\ k_1 \end{bmatrix}$$

choose $k_1 = 1$

$\therefore x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ be the Eigen vector

corresponding to $\lambda = 5$

case (ii): put $\lambda = -2$ in eqn $\textcircled{3}$ & $\textcircled{4}$, we get

$$3x_1 + 4x_2 = 0 \rightarrow \textcircled{7}$$

$$3x_1 + 4x_2 = 0 \rightarrow \textcircled{8}$$

Here $\textcircled{7}$ & $\textcircled{8}$ are same

$$\text{So, } 3x_1 = -4x_2$$

$$\frac{x_1}{x_2} = \frac{-4}{3} = k_2 \text{ (say)}$$

$$x_2 = \begin{bmatrix} -4k_2 \\ 3k_2 \end{bmatrix}$$

choose $k_2 = 1$

$\therefore x_2 = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ be the Eigen vector corresponding to $\lambda = -2$

$$\text{Now, } Q = \frac{1}{\sqrt{17}} \begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}$$

$$Q^{-1} = \frac{1}{|Q|} [\text{adj } Q]$$

$$= \frac{1}{3+4} \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix}$$

which is an invertible matrix

$$\text{Also, } D = Q^{-1} A Q = \frac{1}{7} \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & -4 \\ 3 & 2 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1+4 & -4+12 \\ 3+2 & -12+6 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 5 & 8 \\ 5 & -6 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 15+20 & 24-24 \\ -5+5 & -8-6 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 35 & 0 \\ 0 & -14 \end{bmatrix}$$

which is a required diagonal matrix having eigen values on the diagonal.

over

1) Test the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ for

diagonalizability. If yes, find an invertible matrix P and diagonal matrix D .

(or)

Let T on $\mathbb{R}^2$ defined as $T(a,b) = (5a+4b, a+2b)$ be a linear transformation.

Check for diagonalizability of T , and find basis β such that $[T]_{\beta}$ is a diagonal matrix.

Solution:

Given: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$T(a,b) = (5a+4b, a+2b)$ is $T = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

$\beta = \{(1,0), (0,1)\} \in \mathbb{R}^2$ be the standard basis.

To find the Eigen values & Eigen vectors.

$T(u) = T(1,0) = (5(1)+4(0), 1+2(0)) = (5,1)$

$\Rightarrow (5,1) = 5(1,0) + 1(0,1)$

$M(u) = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \rightarrow \text{Eigen vector}$

$T(v) = T(0,1) = (5(0)+4(1), 0+2(1)) = (4,2)$

$\Rightarrow (4,2) = 4(1,0) + 2(0,1)$

$M(v) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$

By Q. 1, we get

$A = M[T] = [T]_{\beta} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

The characteristic equation is

$\lambda^2 - 7\lambda + 6 = 0$

$g_1 = 5+2 = 7$

$g_2 = 1+2 = 3$

$\therefore$ The characteristic equation is

$\lambda^2 - 7\lambda + 6 = 0$

$(\lambda-1)(\lambda-6) = 0$

and $\lambda = 1, 6$ are the Eigen values of A

Here, multiplicity of $\lambda = 1$ is 1 and of $\lambda = 6$ is 1

(i.e) $\dim(E_1) = 1$

& multiplicity of $\lambda = 6$ is 1

(i.e) $\dim(E_6) = 1$

By Result

$[T]_{\beta} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ is diagonalizable.

To find Eigen vectors

Let $\beta = \{(1,0), (0,1)\}$

$$M(v_0) = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \rightarrow \textcircled{2}$$

By ①, ② we get

$$A = M[T] = [T]^{-1} \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

The characteristic equation is

$$\lambda^2 - 8\lambda + 8 = 0$$

$$g_1 = 5 + 2 = 7$$

$$g_2 = |A| = \begin{vmatrix} 5 & 4 \\ 1 & 2 \end{vmatrix} = 10 - 4 = 6$$

$\therefore$ The characteristic equation is

$$\lambda^2 - 7\lambda + 6 = 0$$

$$(\lambda - 1)(\lambda - 6) = 0$$

$$\lambda = 1, 6$$

are the Eigen values of

$$A$$

Here, multiplicity of 1 is 1 and of 6 is 1

$$\text{e.g. dim } (E_1) = 1, \text{ dim } (E_6) = 1$$

By Result

$$A = M[T]^{-1} \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} M$$

Find Eigen vectors

$$(A - \lambda I)x = 0$$

$$\Rightarrow \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow (5-\lambda)x_1 + 4x_2 = 0 \rightarrow \textcircled{5}$$

$$\Rightarrow (1x_1 + (2-\lambda)x_2 = 0 \rightarrow \textcircled{4}$$

case (I): put $\lambda = 1$ in equ ③ & ④, we get

$$4x_1 + 4x_2 = 0 \rightarrow \textcircled{5}$$

$$1x_1 + 1x_2 = 0 \rightarrow \textcircled{4}$$

Here ③ & ④ are same

$$\text{so, } 4x_1 = -4x_2$$

$$\frac{x_1}{x_2} = \frac{-4x_2}{4} = -1 \text{ (say)}$$

$$x_1 = -x_2$$

choose $k_1 = 1$ be the Eigen vector corresponding to $\lambda = 1$

case (II): put $\lambda = 6$ in equ ③ & ④, we get

$$-1x_1 + 4x_2 = 0 \rightarrow \textcircled{7}$$

$$1x_1 - 4x_2 = 0 \rightarrow \textcircled{8}$$

Here ⑦ & ⑧ are same

$$x_1 = 4x_2$$

$$\frac{x_1}{x_2} = 4 \text{ (say)}$$

Now, $A = \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} [adj A]$

$= \frac{1}{-1-4} \begin{bmatrix} 1 & -4 \\ -1 & -1 \end{bmatrix} = \frac{1}{-5} \begin{bmatrix} 1 & -4 \\ -1 & -1 \end{bmatrix}$

which is an invertible matrix

Also, $D = A^{-1} A Q = \frac{1}{-5} \begin{bmatrix} 1 & -4 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 5 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$= \frac{1}{-5} \begin{bmatrix} 1 & -4 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -5+4 & 5-4 \\ -1+1 & 4-1 \end{bmatrix}$

$= \frac{1}{-5} \begin{bmatrix} 1 & -4 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$= \frac{1}{-5} \begin{bmatrix} -1-4 & 1-4 \\ 1-1 & -1+1 \end{bmatrix}$

$= \frac{1}{-5} \begin{bmatrix} -5 & -3 \\ 0 & -3 \end{bmatrix}$

$D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ which is a regular

diagonal matrix having eigen values are in diagonal.

UNIT II Diagonalization

Summary

1. A square matrix is diagonalizable if and only if it has a full set of linearly independent eigenvectors.

2. The eigenvalues of a diagonalizable matrix are the diagonal elements of the diagonal matrix.

A^n * Matrix of A^n [T]_p

A^n * Eigen values & Eigen vectors

A^n * Eigen values & Eigen vectors - Ordered Basis.

* - Diagonal matrix.

A^n * Diagonalization.

UNIT II Inner Product Space

Definition: Let V be a vector space over F .

and inner product on V is a function

that assigns to every ordered pair

of vectors x and y in V , a scalar

in F , denoted by $\langle x, y \rangle$, such that

for all $x, y, z \in V$ & $\alpha \in F$, such

that the following properties are holds

i) $\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$

ii) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$

iii) $\langle x, y \rangle = \overline{\langle y, x \rangle}$ where $\overline{\langle x, y \rangle}$ is a complex conjugate